



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

**Faculty of
Computing**

SECJ3323-03 –Software Design And Architecture

Medicine Tracking and Refill System
SRS

SECTION : 03
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CASE STUDY TITLE

Medicine Identifier and Refill System

BACKGROUND

The Medicine Identifier and Refill System is a mobile application designed to assist patients, especially the elderly and those with chronic illnesses who often struggle to manage multiple medications, remember dosages, and refill prescriptions on time. The project aims to solve these problems by providing a convenient, intelligent, and secure medicine management platform. Using image recognition technology, users can scan medicine packaging to obtain the medicine's accurate details. Integrated with Stripe for secure payment and Google Maps API, the app enables users to locate nearby pharmacies and order refills online. The development follows an agile methodology, using React Native for cross-platform support, Supabase for backend management, and AI-powered image recognition with OpenCV and TensorFlow Lite. The system also includes an analytical dashboard to track medication history and share reports with healthcare professionals, supporting SDG 3: Good Health and Well-being by promoting accessible and reliable healthcare for all.

PROBLEM STATEMENT

1. Many patients, especially the elderly and chronically ill, find it difficult to identify medicines correctly, leading to medication errors and missed doses. According to the World Health Organization (WHO), medication errors cause at least one death every day and injure approximately 1.3 million people annually.
2. Patients often forget their medication schedules or delay refills, resulting in poor treatment adherence. It is a common issue among older adults managing multiple prescriptions.
3. Limited awareness of side effects and drug interactions increases the risk of adverse reactions, especially when patients self-manage without proper guidance.
4. Unavailable or out-of-stock medicines leave patients without immediate alternatives, disrupting treatment continuity.
5. Current apps lack integration of real-time pharmacy availability and location-based services, forcing patients to manually locate pharmacies.
6. There is a lack of a comprehensive, AI-based system that integrates medicine recognition, refill management, pharmacy locator, and analytics.
7. Patients and doctors lack digital tools to monitor medication history for personalized treatment evaluation.

PROJECT DOMAIN AND COMMUNITY CONTEXT

Domain: Healthcare Technology and Digital Health Innovation

This project engages patients, caregivers, and healthcare professionals to promote safer and more informed medication management. During the development process, the team will gather feedback from elderly users and healthcare workers to ensure usability and accessibility. The system encourages community health awareness through educational features on medicine safety, reminders for adherence, and AI-driven insights that support responsible medication use. By empowering users to manage their health digitally, the project contributes to building a more health-conscious and connected community.

GOAL

To develop a smart, user-friendly mobile application that enhances medication safety and treatment adherence by using AI-based medicine recognition, real-time pharmacy integration, and automated refill management to support safer and more efficient healthcare access.

OBJECTIVES

1. To implement image recognition technology that accurately identifies medicines using medicine packaging scans.
2. To provide detailed information about medicine dosage, side effects, and purpose for patient awareness.
3. To create a medication reminder and refill notification system.
4. To integrate a secure payment gateway for online refill ordering and pickup.
5. To enable users to locate nearby pharmacies.
6. To build an analytical dashboard for tracking medicine intake history and exporting reports.

STAKEHOLDERS

1. Patients
2. Elderly users
3. Chronic illness patients
4. Pharmacies